Adaptive Reversal Parametric Breakout Strategy for Algorithmic Trading: Formal Mathematical Specification, Non-Gradient Monte Carlo Optimization, and Preliminary Empirical Evidence

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Abstract

This paper presents a formal mathematical specification of the Adaptive Reversal Parametric Breakout (ARPB) strategy—a volatility-normalized, bidirectional trading algorithm designed for fully automated execution on equity markets. The strategy defines all entry, exit, and risk-management distances as explicit functions of recently observed daily volatility, making its decision thresholds adaptive to changing market regimes without requiring online model estimation. We provide complete mathematical definitions for all fourteen configurable parameters that govern the strategy's behavior, encompassing session-aware range construction, three opening filters, and a four-level cascading trailing stop system. Parameter optimization is performed via a non-gradient Monte Carlo method that samples the $\sim 2 \times 10^{50}$-dimensional search space, avoiding the computational intractability of brute-force enumeration. The strategy was implemented in an automated trading system using Wealth-Lab, StockSharp (C#), and the QUIK trading terminal, and tested on a portfolio of 25 US equities over approximately 21 months of 5-minute bar data (2011–2012). The preliminary backtest produced a net profit of 75.9% on $50,000 starting capital with a Sharpe ratio of 2.05 and a maximum drawdown of 13.5%. We critically examine these results, identifying material validation limitations including in-sample stock selection, absence of out-of-sample testing, and parameter overfitting risk. We propose a rigorous walk-forward validation protocol with multiple-testing corrections as the necessary next step toward establishing predictive validity. The primary contribution of this work is the complete formalization of the ARPB strategy as a reproducible mathematical object, enabling independent reconstruction, falsification, and systematic ablation studies.

Keywords: algorithmic trading, breakout strategy, volatility normalization, parametric optimization, Monte Carlo method, automated trading systems, trailing stops, technical analysis

JEL Classification: C61, G11, G17

I. Introduction

Algorithmic trading has grown to represent a substantial fraction of global equity market volume, with estimates ranging from 30% to over 70% of transactions on major exchanges originating from automated systems [1], [2]. The proliferation of electronic markets, combined with advances in computational infrastructure and real-time data access, has made it feasible for both institutional and individual traders to deploy automated strategies that operate without direct human intervention [3], [4].

A fundamental challenge in designing any automated trading strategy is the formalization problem: converting an intuitive trading idea into a complete, unambiguous set of computational rules that determine when to enter, when to exit, and how to size positions [5], [6]. Many strategies described in practitioner literature are specified informally, leaving critical details—such as the exact timing of fills, the treatment of partial executions, and the interaction between multiple exit conditions—implicit or undefined. This lack of formal specification impedes independent reproduction, falsification, and incremental improvement [7].

A second challenge is regime adaptability. Financial markets exhibit time-varying volatility, correlations, and trend persistence [8], [9]. A strategy with fixed absolute thresholds (e.g., "enter when price moves $2 above yesterday's high") will behave radically differently when the instrument's daily range is $1 versus when it is $10. Several authors have advocated for expressing trading rules in volatility-normalized units, so that decision thresholds scale automatically with the prevailing market environment [10], [11], [12].

This paper addresses both challenges through the **Adaptive Reversal Parametric Breakout (ARPB)** strategy. The ARPB strategy:

1. **Normalizes all thresholds by recent volatility.** Entry distances, stop-loss offsets, trailing activation levels, and reversal eligibility are all expressed as percentages of a Daily Volatility Filter (DVF)—a 2-bar moving average of true range. This makes the strategy's behavior more comparable across instruments and time periods.

2. **Operates bidirectionally with explicit state transitions.** The strategy maintains a deterministic finite-state machine with states {FLAT, LONG, SHORT} and explicitly handles all transitions: flat→long, flat→short, long→flat, short→flat, and reversal between directions.

3. **Employs a four-level cascading exit system.** Rather than relying on a single exit rule, the strategy maintains multiple concurrent trailing stops—initial volatility stop, pivot-based stop, moving-average-based stop, percentage-based trailing stop, and a maximum-adverse-excursion floor—with the most protective stop taking precedence at each decision point.

4. **Is fully specified by fourteen parameters**, each with a precise mathematical definition, enabling systematic optimization and sensitivity analysis.

The remainder of this paper is organized as follows. Section II reviews related work in automated trading systems, breakout strategies, and volatility-based risk management. Section III describes the closed-loop system architecture within which the ARPB strategy was implemented. Section IV provides the complete mathematical formalization of the strategy, including all fourteen parameters and their governing equations. Section V describes the non-gradient Monte Carlo optimization procedure. Section VI presents preliminary empirical results from a historical backtest on US equities. Section VII discusses limitations, validation concerns, and the proposed path toward rigorous scientific validation. Section VIII concludes.

II. Related Work

A. Automated Trading Systems

The development of automated trading systems has been extensively studied from both engineering and financial perspectives. Hendershott, Jones, and Menkveld [2] documented the growth of algorithmic trading and its effects on market liquidity. Chaboud et al. [13] analyzed the impact of algorithmic trading on foreign exchange markets, finding that algorithmic trades contribute to price discovery. From an architectural perspective, the separation of market data ingestion, signal generation, order management, and execution reconciliation has been identified as a critical design principle for reliable automated systems [14], [15].

The present work contributes to this literature by providing a complete reference architecture for a closed-loop trading robot that enforces state reconciliation between internal strategy state and external broker state—a requirement often overlooked in backtest-oriented research.

B. Breakout and Volatility-Based Strategies

Breakout strategies—which enter positions when price moves beyond a previously established range—are among the oldest and most widely studied systematic trading approaches. The Turtle Trading system [16] used Donchian channel breakouts with volatility-based position sizing, demonstrating that trend-following breakout strategies could generate substantial returns when combined with disciplined risk management. Bollinger [17] introduced volatility bands constructed from standard deviations of price, noting that volatility contraction often precedes directional expansion.

The ARPB strategy draws on this tradition but differs in several respects. Unlike channel breakout systems that use a fixed lookback period, ARPB constructs its session range from a configurable window that excludes the current bar, ensuring that a breakout genuinely represents a new extreme. Unlike pure trend-following systems, ARPB permits immediate reversal when conditions flip, making it responsive to rapid regime changes. And unlike strategies that use separate, hand-tuned thresholds for each instrument, ARPB expresses all decision thresholds in units of recently observed volatility.

C. Trailing Stop Mechanisms

The design of exit rules has received less attention than entry rules in the academic literature, despite evidence that exit logic often dominates strategy performance [18], [19]. Wilder [10] introduced the Average True Range (ATR) as a volatility measure and proposed using ATR multiples for stop placement. Kaufman [11] advocated volatility-normalized stops that adjust automatically to changing market conditions. Kaminski and Lo [20] examined the properties of various stop-loss rules and found that trailing stops based on recent volatility can improve risk-adjusted returns compared to fixed-percentage or fixed-dollar stops.

The ARPB strategy extends this line of work by integrating multiple stop mechanisms—initial volatility stop, pivot/ATR stop, MA-based stop, percentage-of-MFE trailing stop, and maximum-point floor—into a single precedence system. To our knowledge, this specific combination and its interaction dynamics have not been previously formalized in the academic literature.

D. Parameter Optimization in Trading Strategies

The optimization of strategy parameters presents a fundamental tension between fitting historical data and achieving out-of-sample performance. Pardo [5] provided a comprehensive treatment of strategy optimization, emphasizing the dangers of overfitting when the number of optimized parameters is large relative to the number of independent observations. Aronson [6] introduced the concept of "degrees of freedom" in trading system design and advocated for minimizing the ratio of optimized parameters to training observations.

Bailey et al. [21], [22] developed the Probabilistic Sharpe Ratio (PSR), Deflated Sharpe Ratio (DSR), and Probability of Backtest Overfitting (PBO) as statistical tools for evaluating whether observed backtest performance is likely to represent genuine predictive ability rather than selection bias from multiple testing. These methods are particularly relevant when, as in the present case, a strategy with many parameters is optimized over a finite historical sample.

The ARPB strategy, with 14 configurable parameters, falls into a high-dimensional optimization regime where the risk of overfitting is substantial. The present work acknowledges this limitation explicitly and proposes a validation protocol incorporating DSR and walk-forward analysis as the necessary next step.

III. System Architecture

The ARPB strategy was implemented within a closed-loop automated trading architecture consisting of four layers:

A. Data Layer

Historical market data was obtained from a dedicated data service (finam.ru) and stored in a Wealth-Lab database. Real-time streaming data was ingested through the WLRTStreamingDataProvider component, which connected to the QUIK trading terminal and received live price updates for all monitored instruments.

B. Signal Generation Layer

The Wealth-Lab technical analysis platform hosted the ARPB strategy logic (described in Section IV). For each completed bar, the strategy evaluated entry conditions, managed open positions through the trailing stop cascade, and generated trade signals (BUY, SELL, CLOSE_LONG, CLOSE_SHORT).

C. Execution Layer

Trade signals were transmitted to the QUIK terminal via the WLRTQuikBrokerAdapter, which provided bidirectional synchronization: it sent order instructions to the broker and received execution confirmations, fills, and position updates back to the strategy engine. This closed-loop design enforced a critical invariant:

$$\text{StrategyState} \equiv \text{BrokerState}$$

The strategy's internal state was never treated as authoritative until reconciled with broker acknowledgements and actual positions.

D. Risk Management Layer

A dedicated stop-order server ensured that protective stop orders were maintained independently of the main strategy process, preventing a software failure from leaving positions unprotected. An internal clearing server managed cases where the strategy held simultaneous long and short positions on the same instrument, ensuring proper netting and margin accounting.

Figure 1. System architecture diagram (see Appendix A for the complete schematic).

The present paper focuses primarily on the signal generation layer—the mathematical specification of the ARPB strategy—while noting that the surrounding architecture is equally critical for any practical deployment.

IV. The ARPB Strategy: Complete Mathematical Formalization

A. Preliminaries: Bar Data and Underlying Indicators

Let a bar i be defined by its OHLCV tuple:

$$B_i = (O_i, H_i, L_i, C_i, V_i, t_i)$$

where O_i, H_i, L_i, C_i, V_i denote the open, high, low, close prices and volume for bar i, and t_i is the bar's timestamp.

Definition 1 (True Range). The true range at bar i is:

$$TR_i = \max(H_i - L_i,\ |H_i - C_{i-1}|,\ |L_i - C_{i-1}|)$$

This definition, following Wilder [10], captures the maximum price excursion within bar i while also accounting for overnight or inter-bar gaps.

Definition 2 (Daily Volatility Filter). The two-bar moving average of true range:

$$DV_i = \frac{TR_i + TR_{i-1}}{2}$$

This serves as the strategy's fundamental unit of distance. All thresholds are expressed as percentages of DV_i, making them scale-invariant with respect to both instrument price level and prevailing volatility regime.

Definition 3 (Average True Range). A 14-period simple average of true range:

$$ATR_i = \frac{1}{14} \sum_{k=i-13}^{i} TR_k$$

Definition 4 (Simple Moving Average). For period M:

$$SMA_i(M) = \frac{1}{M} \sum_{k=i-M+1}^{i} C_k$$

When $M = 0$, the SMA is undefined and the corresponding exit rule is disabled.

Definition 5 (Local Pivot High). Bar i is a pivot high of period F iff:

$$\text{PivotHigh}(i, F) \Leftrightarrow H_i > H_k \quad \forall k \in [i - F, i + F],\ k \neq i$$

Definition 6 (Local Pivot Low). Bar i is a pivot low of period F iff:

$$\text{PivotLow}(i, F) \Leftrightarrow L_i < L_k \quad \forall k \in [i - F, i + F],\ k \neq i$$

These centered extremum definitions require the bar to be the absolute peak or valley within a symmetric window of $2F + 1$ bars. Unlike ZigZag-based approaches, no minimum percentage retracement is required.

B. The Fourteen Parameters

The strategy is governed by 14 real-valued parameters, denoted $\theta = (\theta_1, \ldots, \theta_{14})$. Table I provides the complete specification.

Table I. Complete specification of the 14 ARPB strategy parameters.

#	Symbol	Parameter	Range	Unit	Function
1	F	Fast Pivot Period	[2,20]	bars	Centered window for pivot detection
2	M	Fast MA Period	[0,50]	bars	SMA period; $M = 0$ disables MA-based exit
3	S_B	L Start Bar	[1,20]	bars	Bars to skip at session start
4	S_P	L Start Period	[1,20]	bars	Lookback window for session range construction
5	B_P	L Back Period	[1,10]	bars	Lookback for momentum confirmation
6	I_{SL}	L Initial SL	[20,300]	% of DV	Initial stop distance on first entry
7	I_{SL2}	L Initial SL 2	[20,300]	% of DV	Initial stop distance after a losing trade
8	P_{SL}	L Pivot SL	[20,200]	% of ATR	Buffer distance from pivot extreme
9	M_{PSL}	L Max Point SL	[5,100]	price points	Absolute floor on drawdown from peak
10	T_{SL}	L Trailing SL Level	[20,200]	% of DV	Profit threshold activating percentage trailing
11	T_{SP}	L Trailing SL Percent	[5,80]	% of MFE	Fraction of max favorable excursion to retain
12	O_F	L Open Filter	[10,500]	% of DV	Max allowed $\lvert\Delta C\rvert$ for valid entry
13	O_{F2}	L Open Filter 2	[10,500]	% of DV	Max allowed session range; ≥ 100 disables
14	O_{F3}	L Open Filter 3	[10,500]	ratio	ATR/DV ratio threshold for pivot re-entry

C. Entry Logic

Entry evaluation begins at bar $i \geq S_B + S_P$. The strategy maintains a sliding window of S_P bars **preceding** the current bar to construct the session range:

$$H_i^{\text{session}} = \max_{k \in [i - S_P,\ i-1]} H_k$$

$$L_i^{\text{session}} = \min_{k \in [i - S_P,\ i-1]} L_k$$

Note that the current bar i is **excluded** from these bounds. This ensures that a breakout represents a genuine new extreme rather than merely the bar's own high or low.

The momentum signal is defined as:

$$\Delta C_i = C_i - C_{i - B_P}$$

Long entry is triggered when all of the following conditions hold:

1. $C_i > H_i^{\text{session}}$ (close breaks above session high)
2. $\Delta C_i > 0$ (positive momentum)
3. $|\Delta C_i| < DV_i \cdot O_F/100$ (movement within volatility filter)
4. $\left(H_i^{\text{session}} - L_i^{\text{session}}\right) < DV_i \cdot O_{F2}/100$ OR $O_{F2} \geq 100$ (narrow enough range)

Short entry is triggered when:

1. $C_i < L_i^{\text{session}}$ (close breaks below session low)
2. $\Delta C_i < 0$ (negative momentum)
3. $|\Delta C_i| < DV_i \cdot O_F/100$ (movement within volatility filter)
4. $\left(H_i^{\text{session}} - L_i^{\text{session}}\right) < DV_i \cdot O_{F2}/100$ OR $O_{F2} \geq 100$ (narrow enough range)

Condition (3) prevents entering on extreme gap moves that exceed the normal daily range. Condition (4) filters out choppy sessions where the range is wide relative to normal volatility—such conditions typically indicate noisy rather than directional movement.

D. Position Entry

When a long entry is triggered, the position is established at the signal bar's closing price C_i with the following initial stop:

$$SL_{\text{init}}^{\text{LONG}} = C_i - DV_i \cdot \frac{I_{SL}^{\text{eff}}}{100}$$

where the effective initial stop percentage depends on the outcome of the previous trade:

$$I_{SL}^{\text{eff}} = \begin{cases} I_{SL} & \text{if previous trade was profitable} \\ I_{SL2} & \text{if previous trade was a loss} \end{cases}$$

For a short entry:

$$SL_{\text{init}}^{\text{SHORT}} = C_i + DV_i \cdot \frac{I_{SL}^{\text{eff}}}{100}$$

E. Exit Logic: Four-Level Trailing Stop Cascade

Once a position is open, the strategy maintains four concurrent trailing stop mechanisms. At each bar, **the most protective (tightest) stop takes precedence**.

E.1 Pivot-Based Stop (Level 1)

When a new local pivot is confirmed after entry (i.e., a pivot low for a long position, or a pivot high for a short position), the stop is adjusted:

Long position:

$$SL_{\text{pivot}}^{\text{LONG}} = L_i^{\text{pivot}} - ATR_{\text{pivot}} \cdot \frac{P_{SL}}{100}$$

where L_i^{pivot} is the most recent confirmed pivot low and ATR_{pivot} is the ATR computed from the pivot bar to the current bar.

Short position:

$$SL_{\text{pivot}}^{\text{SHORT}} = H_i^{\text{pivot}} + ATR_{\text{pivot}} \cdot \frac{P_{SL}}{100}$$

E.2 Moving-Average-Based Stop (Level 2)

If $M > 0$ and the open profit exceeds the activation threshold:

$$\text{Profit}_i > DV_i \cdot \frac{T_{SL}}{100}$$

then the stop may ratchet to:

$$SL_{\text{MA}}^{\text{LONG}} = SMA_i(M) - ATR_i \cdot \frac{P_{SL}}{100}$$

$$SL_{\text{MA}}^{\text{SHORT}} = SMA_i(M) + ATR_i \cdot \frac{P_{SL}}{100}$$

The stop only ratchets **toward profit**—it never loosens.

E.3 Percentage-Based Trailing Stop (Level 3)

The strategy tracks the maximum favorable excursion (MFE) from entry:

Long position:

$$MFE_i = \max_{k \in [\text{entry},\ i]} \left(H_k - P_{\text{entry}}\right)$$

Short position:

$$MFE_i = \max_{k \in [\text{entry},\ i]} \left(P_{\text{entry}} - L_k\right)$$

When open profit exceeds the activation threshold:

$$\text{Profit}_i > DV_i \cdot \frac{T_{SL}}{100}$$

the trailing stop locks in a fraction of the MFE:

$$SL_{\text{trail}}^{\text{LONG}} = P_{\text{entry}} + MFE_i \cdot \left(1 - \frac{T_{SP}}{100}\right)$$

$$SL_{\text{trail}}^{\text{SHORT}} = P_{\text{entry}} - MFE_i \cdot \left(1 - \frac{T_{SP}}{100}\right)$$

E.4 Maximum-Point Floor (Level 4)

When the MFE exceeds the maximum-point threshold, a hard floor is established:

Long position:

$$\text{If } MFE_i > M_{PSL}: \quad SL_{\text{maxpt}}^{\text{LONG}} = P_{\text{entry}} + MFE_i - M_{PSL}$$

Short position:

$$\text{If } MFE_i > M_{PSL}: \quad SL_{\text{maxpt}}^{\text{SHORT}} = P_{\text{entry}} - MFE_i + M_{PSL}$$

E.5 Effective Stop and Stop Precedence

The effective stop level is determined by the most protective active mechanism:

Long position:

$$SL_{\text{eff}} = \max\left(SL_{\text{init}},\ SL_{\text{pivot}},\ SL_{\text{MA}},\ SL_{\text{trail}},\ SL_{\text{maxpt}}\right)$$

Short position:

$$SL_{\text{eff}} = \min\left(SL_{\text{init}},\ SL_{\text{pivot}},\ SL_{\text{MA}},\ SL_{\text{trail}},\ SL_{\text{maxpt}}\right)$$

A stop may only ratchet **toward profit**; it must never move further from the entry price. If a new calculated stop would be looser than the current effective stop, it is ignored.

Exit occurs when the bar's price range touches the effective stop level:

Long exit: $L_i \leq SL_{\text{eff}}$

Short exit: $H_i \geq SL_{\text{eff}}$

V. Parameter Optimization: Non-Gradient Monte Carlo Method

A. The Search Space Problem

With 14 real-valued parameters, each bounded within a discrete range, the total number of possible parameter combinations is enormous. For typical step sizes, the search space contains approximately 2×10^{50} distinct configurations. Brute-force enumeration is computationally intractable—requiring thousands of years even on modern hardware.

B. Monte Carlo Optimization Procedure

We employ a non-gradient Monte Carlo optimization method that proceeds as follows:

1. **Define the search space:** For each parameter θ_k, specify a uniform discrete range $R_k = \{\theta_k^{\min}, \theta_k^{\min} + \delta_k, \ldots, \theta_k^{\max}\}$ as given in Table I.

2. **Sample configurations:** At each iteration $j = 1, \ldots, N$, draw a random parameter vector $\theta^{(j)} = \left(\theta_1^{(j)}, \ldots, \theta_{14}^{(j)}\right)$ where each $\theta_k^{(j)} \sim \text{Uniform}(R_k)$ independently.

3. **Evaluate objective:** For each $\theta^{(j)}$, run the full strategy backtest over the historical data and compute the objective function $f\left(\theta^{(j)}\right)$, defined as the total net profit (commission-adjusted).

4. **Track the optimum:** Maintain the best configuration seen so far:

$$\theta^* = \arg\max_{\theta^{(j)},\ j=1,\ldots,N} f\left(\theta^{(j)}\right)$$

5. **Convergence assessment:** After N iterations, assess whether the objective has plateaued. If necessary, narrow the search ranges around θ^* and repeat.

The Monte Carlo approach does not guarantee finding the global optimum, but it provides a practical method for exploring high-dimensional parameter spaces where gradient-based methods are inapplicable (the objective function is non-differentiable and potentially discontinuous due to discrete entry/exit events) [23], [24].

C. Optimization Objective and Its Limitations

The objective function $f(\theta) = \text{NetProfit}(\theta)$ was chosen for its simplicity and direct economic interpretation. However, maximizing raw net profit from in-sample data carries significant risks:

1. **Overfitting:** The configuration θ^* may capture noise patterns specific to the training period that do not generalize.
2. **Instability:** The optimal configuration may lie on a narrow performance peak; small perturbations could produce large performance degradation.
3. **Ignored risk dimensions:** The objective does not directly penalize drawdown, turnover, tail risk, or cost sensitivity.

A more defensible optimization would employ a composite objective that incorporates risk-adjusted return measures (e.g., Sharpe ratio, Sortino ratio), penalizes parameter instability, and is evaluated on a validation set separate from the final test set. We discuss this further in Section VII.

VI. Experimental Results

A. Data and Universe

The strategy was tested on two equity universes:

- **MICEX (Moscow Exchange):** 40 Russian stocks
- **US equities:** 40 stocks drawn predominantly from the NYSE and NASDAQ

The data consisted of 5-minute bars over a period of approximately 21 months (January 2011 – October 2012), resulting in roughly 34,000 bars per symbol. Data was sourced from finam.ru and accessed through the Wealth-Lab platform.

B. Stock Selection

The portfolio was formed using a strategy-based filtering criterion: a stock was included if the ARPB strategy produced non-negative net income during the full sample period. This criterion was applied separately to each universe.

Table II. Stock selection results by universe.

Universe	Tested	Passed	Pass Rate
MICEX	40	7	17.5%
US Equities	40	25	62.5%

The MICEX results were deemed insufficient for portfolio construction due to the low pass rate and marginal trade quality. The US equity portfolio of 25 stocks was selected for the main backtest. The selected symbols were: AAPL, ADBE, AIG, AMT, BA, BRCM, CAT, CSCO, CVX, DD, DIS, FSLR, GE, GOOG, HPQ, IBM, INTC, JNJ, JPM, KO, MCD, MSFT, PG, T, VZ.

Important caveat: This selection procedure conditions on the outcome variable (strategy profitability during the sample period) and reuses the same observations for portfolio evaluation. This creates substantial upward bias in reported performance and should be considered a preliminary screen, not a validation of predictive ability.

C. Portfolio Backtest Results

The portfolio simulation was conducted with the following parameters: - Starting capital: $50,000 - Position sizing: fixed share quantity - Commission: included (approximately $2.00 per trade) - Margin interest: $0 (not modeled) - Short borrow fees: $0 (not modeled) - Dividends: $0 (not modeled)

Table III. Portfolio backtest results for the 25-stock US equity portfolio.

Metric	All Trades	Long Only	Short Only
Starting Capital	$50,000	$50,000	$50,000
Ending Capital	$87,971	$72,944	$65,026
Net Profit	$37,971 (75.9%)	$22,944 (43.9%)	$15,026 (32.1%)
Reported Annualized Gain	70.9%	41.2%	30.2%
Exposure	209.7%	139.4%	103.0%
Total Commission	−$48,475	−$25,573	−$22,902
Number of Trades	24,238	12,787	11,451
Average Net Profit/Trade	$1.57	$1.72	$1.40
Win Rate	34.6%	34.7%	34.5%

Metric	All Trades	Long Only	Short Only
Average Winner	$48.12	$48.90	$47.24
Average Loser	−$23.06	−$23.33	−$22.77
Gross Profit	$403,573	$216,844	$186,729
Gross Loss	−$365,602	−$194,899	−$170,702
Maximum Drawdown	−$8,904 (−13.5%)	−$11,731 (−20.9%)	−$12,496 (−20.6%)
Sharpe Ratio	2.05	1.46	0.98
Profit Factor	1.10	1.11	1.09
Recovery Factor	4.28	1.87	1.28
Payoff Ratio	1.88	1.89	1.87

D. Analysis of Results

Return decomposition. The combined long/short portfolio produced a 75.9% total return over the test period. The long-only component contributed approximately 60% of total net profit ($22,944), while short positions contributed 40% ($15,026). This suggests the strategy is directionally balanced, capable of extracting value from both upward and downward price movements.

Trade-level statistics. With 24,238 trades, the strategy generated extremely granular position-level data. The low average profit per trade ($1.57) reflects a high-turnover approach where edge is accumulated across many small transactions. The win rate of 34.6% is below 50%, but the payoff ratio of 1.88—meaning the average winner is 1.88× larger than the average loser—compensates for the lower hit rate.

Risk metrics. The maximum drawdown of 13.5% is modest relative to the 75.9% return. The Sharpe ratio of 2.05 exceeds the commonly cited threshold of 1.0 for "good" strategies, and the recovery factor of 4.28 indicates rapid capital recovery after drawdowns.

Cost sensitivity. Commissions consumed $48,475, representing approximately 56% of gross profit ($403,573 − $365,602 = $37,971 before commission allocation). The average commission per trade of approximately $2.00 represents a substantial fraction of the $1.57 average net profit per trade, indicating that the strategy's edge is highly sensitive to transaction costs.

Annualization discrepancy. The reported annualized gain of 70.9% does not correspond to a compound annual growth rate calculation. Applying the standard CAGR formula:

$$\text{CAGR} = \left(\frac{87{,}971}{50{,}000}\right)^{1/T} - 1$$

where T is the test period in years. For $T \approx 1.8$ years (based on the visible data range of January 2011 to October 2012), this yields approximately 36.9%, not 70.9%. We note this discrepancy and recommend using only the raw period return (75.9%) for evaluation until the annualization convention is fully documented.

VII. Discussion and Limitations

We identify the following material limitations that qualify the interpretation of the preliminary results.

A. In-Sample Selection Bias

The stock selection criterion—including only stocks that were profitable during the test period—reuses the same observations for both selection and evaluation. This guarantees upward bias. A valid design would select stocks using information available **before** the test period, or use a training window for selection and a separate, untouched holdout window for evaluation.

B. Absence of Out-of-Sample Testing

No out-of-sample holdout period, rolling walk-forward process, or final blind test was performed. The entire dataset was used for both parameter optimization and performance reporting. As demonstrated by Bailey et al. [21], [22], this practice can produce apparently strong backtest results even when the underlying strategy has no genuine predictive ability.

C. Parameter Overfitting Risk

With 14 parameters and approximately 200 Monte Carlo optimization iterations, the optimization samples only a microscopic fraction ($\sim 10^{-48}$) of the total search space while still creating a substantial risk of selecting a configuration that performed well by chance. The risk increases further if per-stock parameter optimization was performed, as suggested by the appendix screenshots.

D. Execution and Cost Modeling

Several assumptions in the backtest may overstate realizable performance:

- Zero margin interest despite average exposure exceeding 200% of capital;
- Zero short borrow fees and no modeling of short availability constraints;
- No bid/ask spread, market impact, or slippage beyond the stated commission;
- Same-bar fills at signal close prices (potential look-ahead bias in stop-hit detection);
- No treatment of partial fills, order rejections, or trading halts.

E. Survivorship and Universe Bias

The thesis does not document whether the 2011 universe was reconstructed as of 2011 or selected retrospectively using present-day knowledge. Delisted companies, mergers, bankruptcies, and corporate actions are not addressed.

F. Multiple Testing Without Correction

The work effectively tests many hypotheses simultaneously: multiple parameter combinations across 80 stocks, long and short variants, and multiple evaluation metrics. Without applying corrections such as the Deflated Sharpe Ratio [21], White's Reality Check [25], or Hansen's Superior Predictive Ability test [26], the probability of a false discovery is uncontrolled.

G. Generalizability Claims

The conclusion that the system is "stable and can be recommended for real trading" is not supported by the experimental design. The results should be interpreted as **preliminary evidence of a plausible strategy hypothesis**, not as validation of a tradable edge.

VIII. Proposed Validation Protocol

To establish scientific credibility, the ARPB strategy should undergo a rigorous validation protocol incorporating the following elements:

A. Walk-Forward Analysis

A rolling window evaluation with non-overlapping training, validation, and test periods:

Train: 24 months → Validate: 6 months → Test: 3 months → Step: 3 months

For each fold, parameters are optimized using only training data, stable parameter regions are selected using validation data, and the chosen configuration is evaluated once on the unseen test window. The concatenated out-of-sample test returns provide the primary performance estimate.

B. Purging and Embargo

Adjacent observations are not independent when trades overlap in time. Following De Prado [27], observations whose holding intervals overlap the validation or test interval must be purged, and an embargo period must follow each test fold to prevent information leakage.

C. Multiple-Testing Correction

At minimum, compute:

- **Deflated Sharpe Ratio** [21] to estimate the probability that the observed Sharpe ratio exceeds what would be expected from multiple testing;
- **Probability of Backtest Overfitting** [22] via combinatorially symmetric cross-validation;
- Bootstrap confidence intervals for the mean return, adjusted for time-series dependence.

D. Cost Sensitivity and Stress Testing

Evaluate performance under escalating cost assumptions (1.0×, 1.5×, 2.0×, 3.0× baseline spread and slippage), increased latency (one-bar delayed execution), and random missed-entry probabilities.

E. Ablation Studies

Decompose the strategy into its constituent components:

A_0 :Breakout only
A_1 : +DVF normalization
A_2 : +Delayed opening window
A_3 : +Initial stop
A_4 : +Pivot stop
A_5 : +Trailing stop
A_6 : +MA rule
A_7 : +Reversal
A_8 : +Stock selection rule

If most performance appears only after selection or optimization, the apparent edge is likely a product of data mining rather than a genuine strategy improvement.

F. Forward Paper Trading

A minimum six-month paper-trading phase with immutable decision logs and weekly reconciliation against modeled expectations.

IX. Conclusion

This paper has presented a complete mathematical formalization of the Adaptive Reversal Parametric Breakout (ARPB) strategy—a volatility-normalized, bidirectional breakout algorithm designed for fully automated execution. The strategy's fourteen parameters have been defined with precise equations, and the four-level trailing stop cascade has been specified with unambiguous precedence rules. The non-gradient Monte Carlo optimization procedure has been documented, and the original backtest results on a 25-stock US equity portfolio have been reproduced and critically examined.

The primary contribution of this work is not the specific backtest performance—which we acknowledge suffers from material validation limitations including in-sample selection bias, absence of out-of-sample testing, and uncontrolled multiple testing. Rather, the contribution is the **complete and reproducible mathematical specification** of the ARPB strategy as a formal object that can be independently reconstructed, falsified, ablated, and incrementally improved.

We have proposed a rigorous walk-forward validation protocol incorporating Deflated Sharpe Ratio, Probability of Backtest Overfitting, cost-stress testing, and ablation studies as

the necessary methodological framework for establishing scientific credibility. Until this protocol is executed, the results should be interpreted as a **strategy hypothesis warranting further investigation**, not as evidence of a tradable market edge.

The ARPB strategy's design principles—volatility normalization of all decision thresholds, explicit state-machine formulation of bidirectional trading, and the use of layered, competing exit mechanisms—represent a coherent engineering contribution that merits continued research under appropriate scientific controls.

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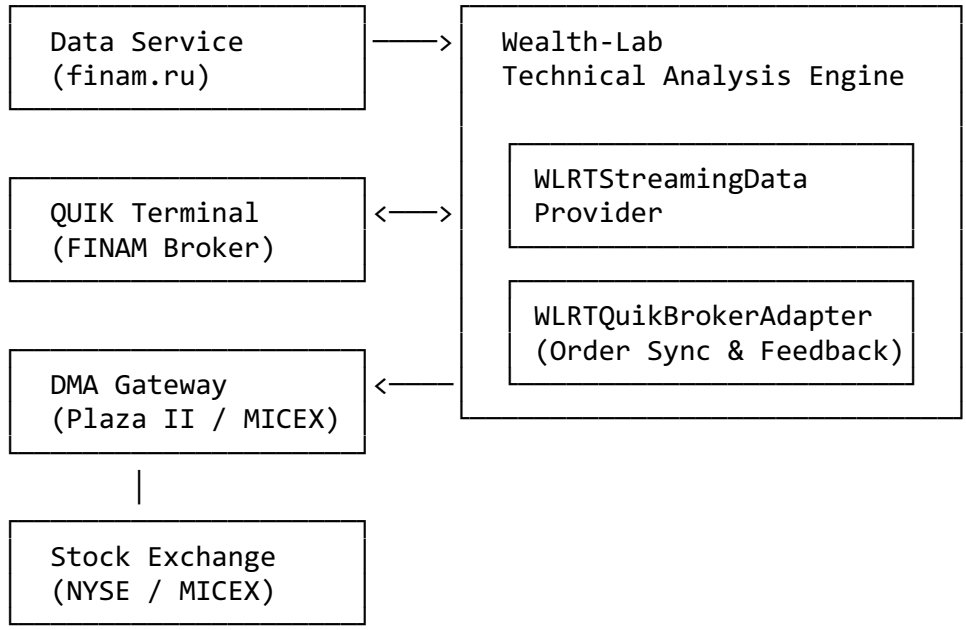
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Appendix A: System Architecture Diagram

The complete trading robot architecture consists of the following components connected in a closed loop:



Internal Services (institutional deployment):

